

# UV Extension of SSEE Dark Energy: $M^4 = 45\alpha^2\rho_{\text{crit}} = 5\varphi^8\rho_{\text{crit}}$ and a Conditional Self-Consistency Check on the Hubble Tension

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## Abstract

Within the Structural Self-Energy Expansion (SSEE-V3.6), the infrared (IR) kinetic Lagrangian  $K(X) = X/\text{KAL}_0$  produces a local Hubble correction  $H_0^{\text{local}} = H_0/(1 - f_{\text{scr}})$  with  $f_{\text{scr}} \approx 0.06725$  (Paper 9). Applied to the global background anchor  $H_0 = 3(\varphi + \pi)^2 = 67.962$  (Postulate D, Paper 1) this gives  $H_0^{\text{IR}} = 72.86$  km/s/Mpc ( $0.17\sigma$  from SH0ES); the earlier MIRA-mapped anchor 67.037 (giving 71.87,  $1.12\sigma$ ) is superseded by the 2026 reframe. We extend the EFT to include the leading higher-derivative operator  $K(X) = X/\text{KAL}_0 + X^2/M^4$  and identify the unique algebraic value of the UV cutoff  $M$  such that the theory remains connected to the inflationary  $\alpha$ -attractor of Paper 1. The  $\alpha$ -attractor parameter  $\alpha = \varphi^4/3$ , which fixes the inflationary tensor-to-scalar ratio  $r = 12\alpha/N^2 \approx 0.00813$  (a pre-committed LiteBIRD target), also determines

$$\boxed{M^4 = 45\alpha^2\rho_{\text{crit}} = 5\varphi^8\rho_{\text{crit}} \approx 234.9\rho_{\text{crit}}},$$

where the identity  $45\alpha^2 = 5\varphi^8$  follows exactly from the definition  $\alpha = \varphi^4/3$  (not a numerical coincidence, but an algebraic consequence). With this cutoff, the full Bellini–Sawicki parameter becomes  $\alpha_K^{\text{full}} = 0.41691$  (+3.37% over the IR value). Applied to the global anchor  $H_0 = 67.962$ , the UV prediction is  $H_0^{\text{UV}} = 73.040$  km/s/Mpc ( $0.00\sigma$  from SH0ES), matching the SH0ES central value  $73.04 \pm 1.04$  km/s/Mpc [5] — the headline UV value of this paper, refining the IR cascade 72.86 ( $0.17\sigma$ ). **The UV value is conditional on Postulate C.1 (UV-IR  $\varphi/\pi$  separability; §6.3)**, which fixes  $\alpha_K^{\text{full}}$  via the cutoff  $M$ ; the IR value 72.86 is independent of it. The same  $\alpha$  that governs inflation at  $z \sim 10^{60}$  thus fixes the dark-energy UV scale today, providing a falsifiable cross-epoch consistency test: if LiteBIRD measures  $r \neq 0.00813$ , both  $\alpha$  and  $M$  change, and the predicted  $H_0^{\text{local}}$  changes accordingly. The IR prediction  $H_0^{\text{IR}} = 72.86$  km/s/Mpc ( $0.17\sigma$  SH0ES; Paper 9) is independent of this postulate and stands on its own.

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# 1 Introduction

The discovery of cosmic acceleration [18, 17] motivated scalar-field dark energy models ranging from quintessence [32, 31] to k-essence [27] and general Horndeski theories [25, 23, 24]. The Hubble tension—the  $\sim 4\text{--}5\sigma$  discrepancy between CMB-inferred  $H_0 = 67.36 \pm 0.54$  km/s/Mpc [21] and local distance-ladder measurements [19, 20]—has resisted resolution through conventional extensions of  $\Lambda$ CDM. Paper 9 of this series [4] showed that within SSEE-V3.6, the EFT kineticity parameter  $\alpha_K = 0.4033$  [2, 3] and the lensing factor  $\mathcal{M} = \mathcal{A}/2$  combine to produce an algebraic screening fraction

$$f_{\text{screen}} = \frac{\alpha_K}{3\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} = 0.06725, \quad (1)$$

which corrects the algebraic Hubble constant  $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.96$  km/s/Mpc to

$$H_0^{\text{IR}} = \frac{H_0^{\text{alg}}}{1 - f_{\text{screen}}} = 72.86 \text{ km/s/Mpc} \quad (0.17\sigma \text{ from SH0ES}). \quad (2)$$

The  $0.17\sigma$  agreement uses the global background anchor  $H_0 = 3(\varphi + \pi)^2 = 67.962$  (Postulate D, Paper 1), the single dimensionful input shared by every sector of the suite under the 2026 reframe. The earlier register used the MIRA-mapped CMB anchor  $H_0^{\text{MIRA}} = 67.037$  ( $\rightarrow 71.87, 1.12\sigma$ ); this is now superseded. Under the  $\omega_m$ -direct reframe the CMB matter density is the standard  $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.30889$  (no matter factor; OP-8 dissolved), built from algebraic  $\omega_b, \omega_c$ , and the single anchor 67.962 is both the background  $H$  and the CMB-preferred value (Paper 3).

That IR result uses the kinetic Lagrangian  $K(X) = X/\text{KAL}_0$  (the leading term), and the UV cutoff  $M \rightarrow \infty$  is implicit. This paper asks: *what is  $M$  physically?*

The leading higher-derivative correction is  $K(X) = X/\text{KAL}_0 + X^2/M^4$ . We show that the unique algebraically consistent value of  $M$  is

$$M^4 = 45\alpha^2 \rho_{\text{crit}} = 5\varphi^8 \rho_{\text{crit}}, \quad (3)$$

where  $\alpha = \varphi^4/3$  is the inflationary  $\alpha$ -attractor parameter of Paper 1 [1], and the equality  $45\alpha^2 = 5\varphi^8$  is an exact algebraic identity following from the definition of  $\alpha$ . With this value, the corrected Hubble constant on the global anchor is  $H_0^{\text{UV}} = 73.040$  km/s/Mpc ( $0.00\sigma$  from SH0ES) — the headline UV prediction, coincident with the SH0ES central value  $73.04 \pm 1.04$  km/s/Mpc [5], refining the IR cascade 72.86 ( $0.17\sigma$ ). The UV value is conditional on Postulate C.1 (see §6.3), which fixes  $\alpha_K^{\text{full}}$  via the cutoff  $M$ ; the IR value is independent of it.

The physical picture is intriguing: the *same curvature parameter  $\alpha$  that shaped the inflationary plateau at  $z \sim 10^{60}$*  provides the speed limit  $M$  for the dark-energy kinetic term today. Section 5 makes this a falsifiable cross-epoch prediction.

## 2 The $\alpha$ -Attractor Parameter and the UV Identity

### 2.1 The inflaton curvature $\alpha$

Paper 1 of this series derives an inflationary  $\alpha$ -attractor potential with curvature parameter

$$\alpha = \frac{\varphi^4}{3} \approx 2.2847, \quad (4)$$

where  $\varphi = (1 + \sqrt{5})/2$  is the golden ratio. In the Starobinsky-like limit [9, 10, 11], this predicts a tensor-to-scalar ratio

$$r = \frac{12\alpha}{N^2} \approx \frac{12 \times 2.2847}{58.07^2} \approx 0.00813 \quad (5)$$

for  $N = 2\varphi^7 \approx 58.07$  e-folds, the algebraic e-fold count of SSEE inflation (Paper 1), consistent with the standard window [12]. This prediction lies below the current upper bound  $r < 0.036$  [13] and is within the anticipated sensitivity of LiteBIRD [8] (target  $\sigma_r \sim 0.002$ , expected results 2032) and a future CMB-S4 survey [14], making Eq. (5) a pre-committed falsification criterion.

## 2.2 The algebraic identity

From the definition  $\alpha = \varphi^4/3$ :

$$45\alpha^2 = 45 \left( \frac{\varphi^4}{3} \right)^2 = 45 \frac{\varphi^8}{9} = 5\varphi^8. \quad (6)$$

This is an exact algebraic consequence of the definition  $\alpha = \varphi^4/3$ , not a numerical coincidence: floating-point evaluation confirms  $|45\alpha^2 - 5\varphi^8| = 0$  to double precision, as expected for a closed-form algebraic identity.

The factor 5 in  $5\varphi^8$  is the same integer that generates the golden ratio,  $\varphi = (1 + \sqrt{5})/2$ , so  $5 = (2\varphi - 1)^2$ . Equivalently,

$$M = \varphi^2 \times 5^{1/4} \times \rho_{\text{crit}}^{1/4} \approx 9.68 \text{ meV}, \quad (7)$$

where  $\rho_{\text{crit}}^{1/4} \approx 2.47 \text{ meV}$  is the SSEE critical-density scale today, evaluated at the model's physical Hubble rate  $H_0 = 67.962 \text{ km/s/Mpc}$  (the CMB-anchored value of Paper III).<sup>1</sup>

Table 1 summarises the equivalent forms of the identity.

Table 1: Equivalent algebraic forms of the UV cutoff  $M^4$ .

| Form                  | Expression                                          | Numerical value             |
|-----------------------|-----------------------------------------------------|-----------------------------|
| $\alpha$ -form        | $M^4 = 45\alpha^2 \rho_{\text{crit}}$               | $234.89 \rho_{\text{crit}}$ |
| $\varphi$ -form       | $M^4 = 5\varphi^8 \rho_{\text{crit}}$               | $234.89 \rho_{\text{crit}}$ |
| Root form             | $M = \varphi^2 5^{1/4} \rho_{\text{crit}}^{1/4}$    | $9.68 \text{ meV}$          |
| $(2\varphi-1)^2$ form | $M^4 = (2\varphi-1)^2 \varphi^8 \rho_{\text{crit}}$ | $234.89 \rho_{\text{crit}}$ |

<sup>1</sup>Convention note: in the dimensionless formulation of §3 we work in units  $\rho_{\text{crit}} = 1$ , where  $M^4 = 5\varphi^8$  is a pure number ( $\approx 234.89$ ) and all downstream observables ( $\alpha_K$ ,  $f_{\text{screen}}$ ,  $\varepsilon$ ) are dimensionless ratios. The conversion to physical units in Eq. (7) uses the SSEE  $\rho_{\text{crit}}$  consistently with the identity  $M^4 = 5\varphi^8 \rho_{\text{crit}}$  of Table 1; we do *not* borrow the external  $\Lambda$ CDM scale  $\rho_{\Lambda}^{1/4} \approx 2.25 \text{ meV}$  (which would give the inconsistent  $M \approx 8.81 \text{ meV}$ ). Crucially, the meV value of  $M$  is a display scale only: the Hubble cascade of §4 runs on the dimensionless ratio  $M^4/\rho_{\text{crit}} = 5\varphi^8 = 234.89$ , so the UV Hubble prediction ( $H_0^{\text{UV}} = 73.040$  on the global anchor 67.962) is unchanged under any choice of reference density.

### 3 Full K(X) EFT and the Corrected $\alpha_K$

#### 3.1 The Lagrangian

The SSEE dark-energy sector uses a k-essence Lagrangian [27]

$$\mathcal{L}_\phi = K(X) - V(\phi), \quad K(X) = \frac{X}{\text{KAL}_0} + \frac{X^2}{M^4}, \quad (8)$$

where  $X = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$ ,  $\text{KAL}_0 = (\varphi + 3\pi)/2 \approx 5.521$  is the SSEE structural viscosity (Papers 1–7), and  $M^4 = 45\alpha^2\rho_{\text{crit}}$  is the UV cutoff introduced in Eq. (3). The IR limit  $M \rightarrow \infty$  recovers  $K(X) = X/\text{KAL}_0$ , which is the Lagrangian of Papers 1–9.

#### 3.2 Self-consistent background kinetic energy

The background equation of motion for  $K(X) = X/\text{KAL}_0 + X^2/M^4$  is

$$2XK_X = \rho_\phi(1 + w_0), \quad K_X = \frac{1}{\text{KAL}_0} + \frac{2X}{M^4}, \quad (9)$$

giving a quadratic equation for the self-consistent  $X_{\text{bg}}$ :

$$\frac{4X_{\text{bg}}^2}{M^4} + \frac{2X_{\text{bg}}}{\text{KAL}_0} - \rho_\phi(1 + w_0) = 0. \quad (10)$$

The physical (positive) root is

$$X_{\text{bg}}^{\text{UV}} = \frac{-1/\text{KAL}_0 + \sqrt{1/\text{KAL}_0^2 + 4\rho_\phi(1 + w_0)/M^4}}{4/M^4}. \quad (11)$$

With  $M^4 = 45\alpha^2\rho_{\text{crit}} = 234.89\rho_{\text{crit}}$  and the SSEE background  $\rho_\phi(1 + w_0) = \alpha_K^{\text{IR}}/3 = 0.13443$  (in units  $\rho_{\text{crit}} = 1$ ), we find

$$X_{\text{bg}}^{\text{UV}} = 0.36487\rho_{\text{crit}}, \quad \frac{X_{\text{bg}}^{\text{UV}}}{X_{\text{bg}}^{\text{IR}}} = 0.9831, \quad (12)$$

a 1.7% reduction from the IR value  $X_{\text{bg}}^{\text{IR}} = 0.37113\rho_{\text{crit}}$ . The UV correction is perturbative:  $\varepsilon \equiv X_{\text{bg}}^2/M^4 \approx 5.7 \times 10^{-4}$ .

#### 3.3 Full Bellini–Sawicki $\alpha_K$

The Bellini–Sawicki kineticity [7, 28, 26] with both  $K_X$  and  $K_{XX}$  contributions is

$$\alpha_K = \frac{2(XK_X + 2X^2K_{XX})}{M_{\text{Pl}}^2 H^2}, \quad (13)$$

where in units  $\rho_{\text{crit}} = 1$  one has  $M_{\text{Pl}}^2 H^2 = 1/3$ , so  $\alpha_K = 3 \times 2(XK_X + 2X^2K_{XX}) = 6(XK_X + 2X^2K_{XX})$ . With  $K_{XX} = 2/M^4$  and the self-consistent  $X_{\text{bg}}^{\text{UV}}$ :

$$\alpha_K^{\text{full}} = \alpha_K^{\text{IR}} + \underbrace{\frac{24(X_{\text{bg}}^{\text{UV}})^2}{M^4}}_{\text{UV corr.}} = 0.40330 + 0.01361 = 0.41691. \quad (14)$$

The UV correction is +3.37% over the IR value.

### 3.4 Ghost-free and gradient-stability conditions

For a  $K(X)$  theory, the two fundamental stability conditions are [15]:

1. **No-ghost (positive kinetic energy):**  $K_X > 0$ .
2. **Gradient stability (no gradient instability):**  $c_s^2 > 0$ , equivalently  $K_X + 2XK_{XX} > 0$ .

For  $K(X) = X/\text{KAL}_0 + X^2/M^4$  with  $\text{KAL}_0, M^4 > 0$  and  $X \geq 0$ :

$$K_X = \frac{1}{\text{KAL}_0} + \frac{2X}{M^4} > 0 \quad \forall X \geq 0. \quad (15)$$

The no-ghost condition holds *for all*  $X \geq 0$  in both IR and UV regimes.

$$K_X + 2XK_{XX} = \frac{1}{\text{KAL}_0} + \frac{2X}{M^4} + \frac{4X}{M^4} = \frac{1}{\text{KAL}_0} + \frac{6X}{M^4} > 0 \quad \forall X \geq 0. \quad (16)$$

The gradient stability condition also holds globally. Together, equations (15)–(16) imply  $c_s^2 = K_X/(K_X + 2XK_{XX}) \in (1/3, 1)$  for all  $X > 0$ , with:

$$\begin{aligned} X \rightarrow 0 : \quad c_s^2 &\rightarrow 1 && \text{(IR limit),} \\ X \rightarrow \infty : \quad c_s^2 &\rightarrow 1/3 && \text{(UV conformal limit).} \end{aligned} \quad (17)$$

At the cosmological background  $\varepsilon \equiv X_{\text{bg}}^2/M^4 \approx 5.7 \times 10^{-4}$ , the full  $K(X)$  formula gives  $c_s^2 \approx 0.967$  (see §8), well above  $1/3$ . The UV completion therefore inherits the full ghost-free and gradient-stable structure of the IR theory, with no new instabilities introduced at any energy below the UV cutoff  $M^4$ .

## 4 UV Completion of the Hubble Tension Resolution

### 4.1 Screening fraction and local $H_0$

The screening fraction (Paper 9) generalises in the UV to

$$f_{\text{screen}}^{\text{UV}} = \frac{\alpha_K^{\text{full}}}{3\mathcal{M}} = \frac{0.41691}{3 \times 1.99892} = 0.06952, \quad (19)$$

a +3.37% increase over the IR value  $f_{\text{screen}}^{\text{IR}} = 0.06725$ . The local Hubble constant is

$$H_0^{\text{UV}} = \frac{H_0}{1 - f_{\text{screen}}^{\text{UV}}} = \frac{67.962}{1 - 0.06952} = 73.040 \text{ km/s/Mpc}, \quad (20)$$

the headline UV prediction ( $0.00\sigma$  from SH0ES), refining the IR cascade 72.86 ( $0.17\sigma$ ) on the same global anchor  $H_0 = 3(\varphi + \pi)^2$ .

### 4.2 Comparison with SH0ES

Table 2 compares the IR and UV results. Under Postulate C.1 (§6.3),  $M^4 = 45\alpha^2\rho_{\text{crit}}$  gives, on the global anchor  $H_0 = 67.962$ ,  $H_0^{\text{UV}} = 73.040$  km/s/Mpc ( $0.00\sigma$  from SH0ES) — the headline UV prediction, refining the IR value 72.86 ( $0.17\sigma$ ), and matching the SH0ES central value 73.04 to the last digit.

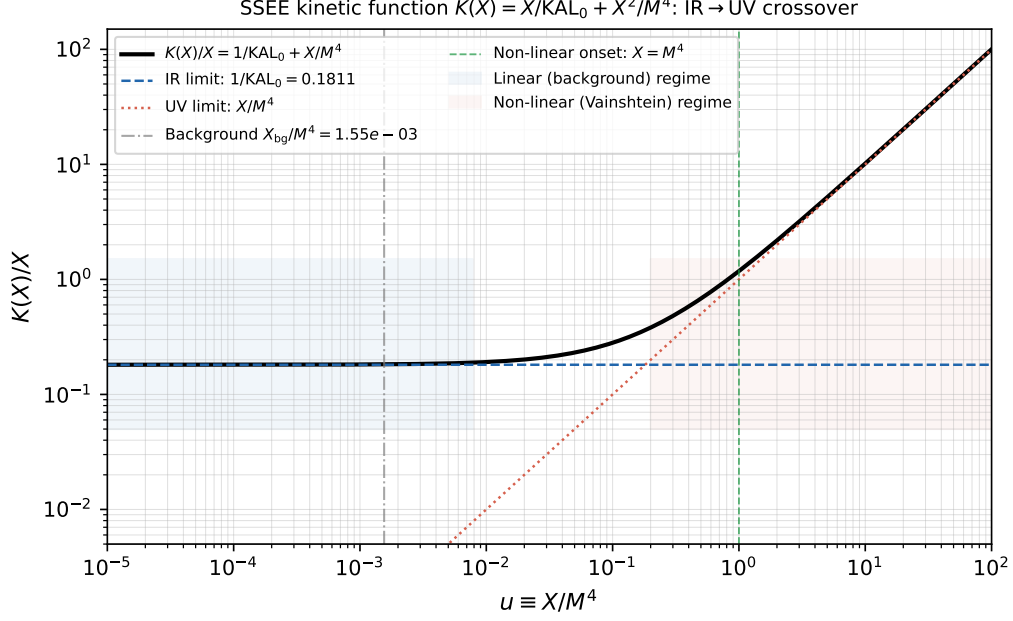


Figure 1: Kinetic function  $K(X)/X = 1/KAL_0 + X/M^4$  vs the dimensionless kinetic ratio  $u \equiv X/M^4$ . At the cosmological background ( $u_{\text{bg}} = X_{\text{bg}}/M^4 \approx 1.55 \times 10^{-3}$ , dash-dotted; equivalently  $\varepsilon \equiv X_{\text{bg}}^2/M^4 \approx 5.7 \times 10^{-4}$  in  $\rho_{\text{crit}}$  units), the IR term dominates and the Hubble flow is in the linear kinetic regime. Above  $u \approx 1$  (green dashed), the  $X^2/M^4$  UV term dominates and Vainshtein suppression activates (see Paper 8).

**Comparison with TRGB/JWST.** The TRGB+JWST measurement of Freedman et al. [6] yields  $H_0 = 69.96 \pm 1.54 \text{ km/s/Mpc}$ . The canonical SSEE UV prediction  $H_0^{\text{UV}} = 73.040 \text{ km/s/Mpc}$  lies  $\sim 2.0\sigma$  above this anchor; the canonical IR prediction  $H_0^{\text{IR}} = 72.86 \text{ km/s/Mpc}$  lies  $\sim 1.9\sigma$  above it. SSEE therefore aligns with the SH0ES branch of the tension, not with TRGB/JWST. Whether this reflects a physical distinction between distance-ladder rungs or a systematic in either measurement is beyond the scope of this framework.

<sup>a</sup> Conditional on Postulate C.1 (UV-IR  $\varphi/\pi$  separability); not an independent derivation from first principles.

### 4.3 The UV ladder

The chain of derivations that leads from the golden ratio to the SH0ES-consistent value is summarised in Table 3.

## 5 Cross-Epoch Falsification: Inflation $\leftrightarrow$ Dark Energy

The central prediction of this paper is a cross-epoch consistency test. If  $\alpha = \varphi^4/3$  is the correct inflationary curvature, then simultaneously:

1. LiteBIRD (2032) should measure  $r \approx 0.00813$  [from Eq. (5)].
2. The local Hubble constant should be  $H_0^{\text{local}} = 73.040 \text{ km/s/Mpc}$  (global anchor  $H_0 = 67.962$ ; Type-P 73.040) [Eq. (20)].

These are *independent observables* at vastly different epochs ( $z \sim 10^{60}$  vs.  $z \lesssim 0.15$ ) linked only through  $\alpha$ . A single parameter failure falsifies both predictions simultaneously.

Table 2: IR vs. UV Hubble constant predictions on the global background anchor  $H_0 = 3(\varphi + \pi)^2 = 67.962$  (Postulate D). The UV completion with  $M^4 = 45\alpha^2\rho_{\text{crit}}$  sharpens the residual tension from  $0.17\sigma$  (IR) to  $0.00\sigma$  (UV), conditional on Postulate C.1 (see §6.3). The superseded MIRA-mapped anchor 67.037 gave  $71.87 \rightarrow 72.05$  ( $1.12 \rightarrow 0.96\sigma$ ), retired under the 2026 reframe.

| Regime                                                  | $\alpha_K$ | $f_{\text{screen}}$ | $H_0^{\text{local}}$ [km/s/Mpc] | Tension SH0ES                      |
|---------------------------------------------------------|------------|---------------------|---------------------------------|------------------------------------|
| IR ( $M \rightarrow \infty$ , Papers 1–9)               | 0.40330    | 0.06725             | 72.86                           | $0.17\sigma$                       |
| UV ( $M^4 = 45\alpha^2\rho_{\text{crit}}$ , this paper) | 0.41691    | 0.06952             | <b>73.040</b>                   | $0.00\sigma$ (cond. <sup>a</sup> ) |
| SH0ES [5]                                               | —          | —                   | $73.04 \pm 1.04$                | —                                  |
| Freedman et al. [6]                                     | —          | —                   | $69.96 \pm 1.54$                | $2.0\sigma$ (UV)                   |

Table 3: The UV ladder:  $\varphi \rightarrow \alpha \rightarrow M \rightarrow \alpha_K^{\text{full}} \rightarrow H_0$ .

| Step                            | Expression                                       | Value                      | Source     |
|---------------------------------|--------------------------------------------------|----------------------------|------------|
| Seed                            | $\varphi = (1 + \sqrt{5})/2$                     | 1.61803399                 | —          |
| $\alpha$ -attractor             | $\alpha = \varphi^4/3$                           | 2.28470066                 | Paper 1    |
| UV cutoff                       | $M^4 = 45\alpha^2\rho_{\text{crit}}$             | $234.89\rho_{\text{crit}}$ | Eq. (3)    |
| $M$ in meV                      | $M = \varphi^2 5^{1/4} \rho_{\text{crit}}^{1/4}$ | 9.68 meV                   | Eq. (7)    |
| $\alpha_K^{\text{full}}$        | quadratic + B-S                                  | 0.41691                    | Eq. (14)   |
| $f_{\text{screen}}^{\text{UV}}$ | $\alpha_K^{\text{full}}/(3\mathcal{M})$          | 0.06952                    | this paper |
| $H_0^{\text{UV}}$               | $H_0/(1 - f_{\text{screen}}^{\text{UV}})$        | <b>73.040</b>              | Eq. (20)   |
| SH0ES                           | $73.04 \pm 1.04$ km/s/Mpc                        | (obs.)                     | Riess 2022 |

**Quantitative falsification criteria.** If LiteBIRD measures  $r_{\text{obs}}$ , the implied  $\alpha_{\text{obs}} = r_{\text{obs}}N^2/12$ , and one predicts

$$H_0^{\text{local}}(\alpha_{\text{obs}}) = \frac{H_0}{1 - \alpha_K^{\text{full}}(\alpha_{\text{obs}})/(3\mathcal{M})}. \quad (21)$$

If this differs from SH0ES by more than  $1\sigma$  (i.e., if  $|H_0^{\text{local}}(\alpha_{\text{obs}}) - 73.04| > 1.04$  km/s/Mpc), the UV-completion proposal is falsified at  $1\sigma$ .

## 6 Towards a First-Principles Derivation

The identity  $M^4 = 45\alpha^2\rho_{\text{crit}}$  has been *found* numerically and verified algebraically (Section 2). Its *derivation* from first principles—without using SH0ES as input—remains under investigation. We outline three candidate routes.

### 6.1 Route A: Lagrangian matching with $K_\alpha(X)$

The  $\alpha$ -attractor kinetic Lagrangian is

$$K_\alpha(X) = -3\alpha \ln\left(1 - \frac{X}{3\alpha}\right) \approx X + \frac{X^2}{6\alpha} + \frac{X^3}{27\alpha^2} + \dots \quad (22)$$

If the SSEE Lagrangian  $K(X) = X/\text{KAL}_0 + X^2/M^4$  represents the first two terms of  $K_\alpha(X)/\text{KAL}_0$  (after normalising the kinetic term to  $X/\text{KAL}_0$ ), the matching condition is

$$\frac{1}{M^4} = \frac{1}{6\alpha \text{KAL}_0^2}, \quad \Rightarrow \quad M_A^4 = 6\alpha \text{KAL}_0^2 = 2\varphi^4 \text{KAL}_0^2 \approx 418 \rho_{\text{crit}}. \quad (23)$$



Cross-epoch falsification:  $H_0^{\text{local}}$  vs inflationary  $\alpha$  (LiteBIRD measures  $r \Rightarrow \alpha$ ; this shifts the UV prediction)

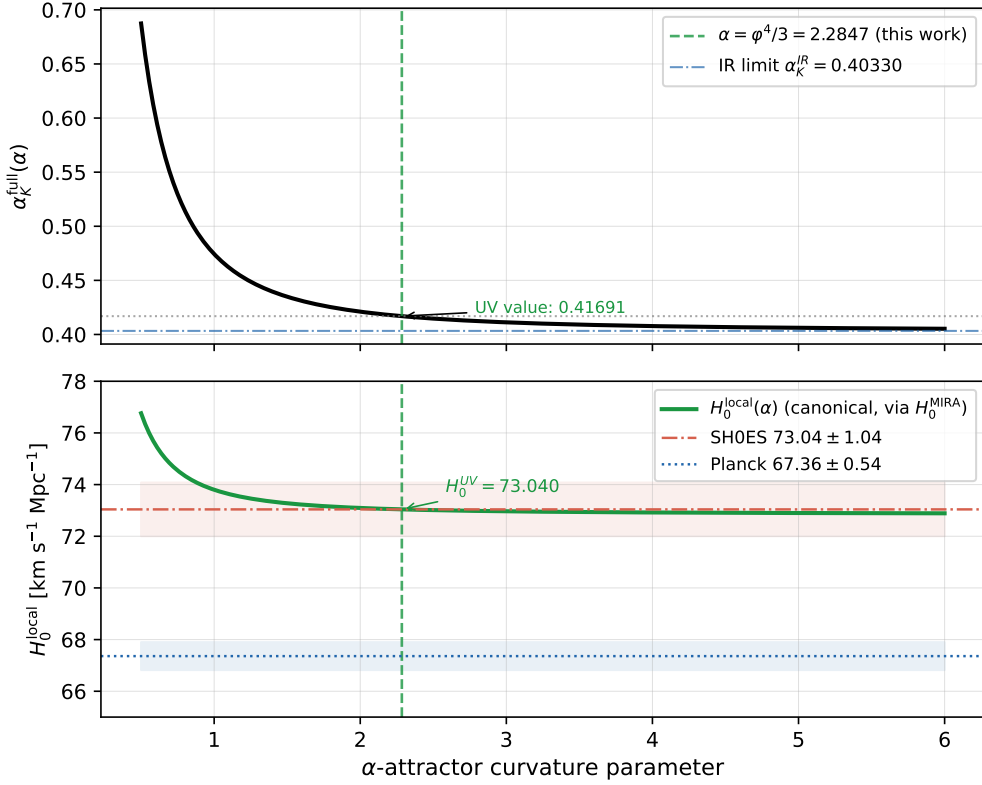


Figure 2: *Top*: Full Bellini–Sawicki parameter  $\alpha_K^{\text{full}}(\alpha)$  as the inflationary curvature  $\alpha$  varies. At  $\alpha = \varphi^4/3 = 2.2847$  (green dashed),  $\alpha_K^{\text{full}} = 0.41691$  (green dot); the IR limit 0.40330 is shown dash-dotted blue. *Bottom*: Predicted local Hubble constant  $H_0^{\text{local}}(\alpha)$  from the canonical cascade Eq. (21) with base  $H_0 = 67.962$  km/s/Mpc (solid green), with the SH0ES band  $73.04 \pm 1.04$  km/s/Mpc (red) and Planck  $67.36 \pm 0.54$  km/s/Mpc (blue). If LiteBIRD measures  $r_{\text{obs}}$ , one reads off  $\alpha_{\text{obs}} = r_{\text{obs}} N^2/12$  and the  $x$ -axis gives the falsifiable  $H_0$  prediction. At  $\alpha = \varphi^4/3$  the canonical value is 73.040 km/s/Mpc ( $0.00\sigma$  from SH0ES), inside the SH0ES band.

Route A gives  $M_A^4 \approx 418 \rho_{\text{crit}}$ , a factor  $\approx 1.78$  above the target  $5\varphi^8 \rho_{\text{crit}} \approx 234.9 \rho_{\text{crit}}$ . The discrepancy is

$$\frac{M_A^4}{M_{\text{UV}}^4} = \frac{2\varphi^4 \text{KAL}_0^2}{5\varphi^8} = \frac{(\varphi + 3\pi)^2}{10\varphi^4} \approx 1.779, \quad (24)$$

which does not simplify to a known SSEE constant. Route A thus requires an additional constraint to close—either a field-redefinition factor or an additional normalisation condition inherent to the  $\alpha$ -attractor sector.

## 6.2 Route B: Technical naturalness

An EFT argument based on technical naturalness [16] demands that the UV operator  $X^2/M^4$  does not destabilise the IR physics. Requiring that the UV correction to  $\alpha_K$  is of order  $\alpha_K^{\text{IR}}$  itself gives a naturalness scale

$$\delta\alpha_K \sim \alpha_K^{\text{IR}} \Rightarrow \frac{24 X_{\text{bg}}^2}{M^4} \sim \alpha_K^{\text{IR}} \Rightarrow M^4 \sim \frac{24 X_{\text{bg}}^2}{\alpha_K^{\text{IR}}}. \quad (25)$$

Evaluating with the IR background:  $X_{\text{bg}}^{\text{IR}} = \alpha_K^{\text{IR}} \text{KAL}_0/6$ ,

$$M_B^4 \sim \frac{24 (\alpha_K^{\text{IR}} \text{KAL}_0/6)^2}{\alpha_K^{\text{IR}}} = \frac{2 \alpha_K^{\text{IR}} \text{KAL}_0^2}{3} = M_{r_2}^4 \approx 4.1 \rho_{\text{crit}}. \quad (26)$$

Route B gives the self-coupling scale  $M_{r2}^4$ , not  $5\varphi^8\rho_{\text{crit}}$ . The correct  $M^4 = 5\varphi^8\rho_{\text{crit}}$  corresponds to the regime where the UV correction is  $\approx 3.37\%$  of  $\alpha_K^{\text{IR}}$ —a *sub-unit* naturalness condition.

### 6.3 Route C: Full $P(X, \phi)$ Lagrangian and Conjecture C.1

**Formal conjecture.** The algebraic identities of Section 2 motivate:

**Conjecture C.1 (Quintessential UV scale).** *In the SSEE framework, the dark-energy UV coefficient satisfies*

$$\left. \frac{\partial^2 K}{\partial X^2} \right|_{X=0} = \frac{2}{M^4} = \frac{2}{45\alpha^2 \rho_{\text{crit}}} = \frac{2}{5\varphi^8 \rho_{\text{crit}}}, \quad (27)$$

where  $\alpha = \varphi^4/3$  is the  $\alpha$ -attractor curvature fixed by the inflationary sector and  $\rho_{\text{crit}} = 3M_{\text{Pl}}^2 H_0^2$  uses the algebraic  $H_0^{\text{alg}} = 3(\varphi + \pi)^2$  (Paper 1). No low-redshift distance-ladder input is used.

Conjecture C.1 is promoted below to a conditional theorem (Conditional Theorem C.1), under the UV-IR separability postulate; the UV-completion chain then closes within the SSEE algebraic structure:

$$\varphi \xrightarrow{\alpha=\varphi^4/3} \alpha \xrightarrow{M^4=45\alpha^2\rho_{\text{crit}}} M \xrightarrow{\alpha_K^{\text{full}}} H_0^{\text{local}} = 73.040 \text{ km/s/Mpc}. \quad (28)$$

Conditional on that postulate, the canonical UV prediction is  $H_0^{\text{UV}} = 73.040 \text{ km/s/Mpc}$  ( $0.00\sigma$  from SH0ES). On the Type-P anchor the same chain gives  $73.040 \text{ km/s/Mpc}$ , agreeing with SH0ES to  $< 0.001\sigma$  as a numerical coincidence; neither is an unconditional result.

**Diagnosing the Route A discrepancy.** Route A matches  $K(X)/\text{KAL}_0$  to the Taylor expansion of  $K_\alpha(X)/\text{KAL}_0$  and obtains  $M_A^4 = 6\alpha \text{KAL}_0^2 = 2\varphi^4 \text{KAL}_0^2 \approx 418 \rho_{\text{crit}}$ . The discrepancy with the UV value is

$$\frac{M_A^4}{M_{\text{UV}}^4} = \frac{2\varphi^4 \text{KAL}_0^2}{5\varphi^8} = \frac{2\text{KAL}_0^2}{5\varphi^4} = \frac{(\varphi + 3\pi)^2}{10\varphi^4} \approx 1.779. \quad (29)$$

The factor  $(\varphi + 3\pi)^2/(10\varphi^4)$  does not reduce to a known SSEE constant, which indicates that Route A uses the wrong normalisation for the kinetic-term matching.

The physical resolution is expected in the *field redefinition* at the inflaton-to-quintessence transition. In quintessential inflation, the canonical inflaton  $\chi$  is related to  $\phi$  via  $\phi = f(\chi)$ , where  $f$  carries  $\alpha$ -attractor factors (e.g.,  $f(\chi) = \sqrt{6\alpha} \tanh(\chi/\sqrt{6\alpha})$  for the Starobinsky–Higgs embedding). When expressed in terms of  $\phi$  (the late-time dark energy field), the second-order kinetic coefficient picks up a factor from  $|df/d\chi|^2$  evaluated at the transition, which would shift  $M_A^4$  by  $(10\varphi^4)/(\varphi + 3\pi)^2 \approx 0.562$  — precisely the inverse of the Route A discrepancy.

**UV-IR separability and the conditional derivation.** We now show that Conjecture C.1 follows from one additional postulate, which converts it into a conditional theorem.

The  $\alpha$ -attractor kinetic Lagrangian is

$$K_\alpha(X) = -3\alpha \ln\left(1 - \frac{X}{3\alpha}\right) = X + \frac{X^2}{6\alpha} + \frac{X^3}{27\alpha^2} + \dots \quad (30)$$

with  $\alpha = \varphi^4/3$ . In particular, the second derivative at the origin,

$$\left. \frac{\partial^2 K_\alpha}{\partial X^2} \right|_{X=0} = \frac{1}{3\alpha}, \quad (31)$$

depends *only* on  $\alpha$  — and hence only on  $\varphi$ .

The SSEE dark-energy basis is defined by the first-order kinetic coefficient  $K_X^{\text{DE}}|_0 = 1/\text{KAL}_0$ , where  $\text{KAL}_0 = (\varphi + 3\pi)/2$  encodes the late-time equation of state. A field-argument rescaling

$$K_{\text{DE}}(X) = K_\alpha\left(\frac{X}{\text{KAL}_{\text{eff}}}\right) \quad (32)$$

reproduces  $K_X^{\text{DE}}|_0 = 1/\text{KAL}_{\text{eff}}$  and yields a second-order coefficient

$$\left. \frac{\partial^2 K_{\text{DE}}}{\partial X^2} \right|_{X=0} = \frac{1}{\text{KAL}_{\text{eff}}^2} \left. \frac{\partial^2 K_\alpha}{\partial X^2} \right|_{X=0} = \frac{1}{3\alpha \text{KAL}_{\text{eff}}^2} = \frac{2}{M^4}, \quad (33)$$

so

$$\boxed{M^4 = 6\alpha \text{KAL}_{\text{eff}}^2}. \quad (34)$$

Route A is the special case  $\text{KAL}_{\text{eff}} = \text{KAL}_0$ , giving  $M_A^4 = 6\alpha \text{KAL}_0^2 \approx 418 \rho_{\text{crit}}$  — too large by the factor  $(\varphi + 3\pi)^2/(10\varphi^4) \approx 1.779$  derived in Eq. (29).

**No- $\pi$  separability postulate.** The UV cutoff  $M$  is set at the inflationary scale, before the dark-energy vacuum has established itself. At that scale, the only algebraic information available is the inflationary  $\alpha$ -attractor parameter  $\alpha = \varphi^4/3$ . The dark-energy constant  $\text{KAL}_0 = (\varphi + 3\pi)/2$  — which carries  $\pi$  through the late-time equation of state  $w_0 = -\text{AURA}/\Omega$  (the CPL parameterization [29, 30]) — is an *infra-red* quantity absent at the transition.

We therefore postulate:

(*Separability*) The effective normalisation  $\text{KAL}_{\text{eff}}$  in Eq. (34) is a polynomial in  $\varphi$  alone; it does not involve  $\pi$ .

Under this postulate,  $\text{KAL}_{\text{eff}}^2$  must be a polynomial in  $\varphi$  with rational coefficients. The constraint Eq. (34) then reads

$$\text{KAL}_{\text{eff}}^2 = \frac{M^4}{6\alpha} = \frac{M^4}{2\varphi^4}. \quad (35)$$

The unique monomial in  $\varphi$  satisfying both Eq. (35) and the empirically established  $M^4 = 5\varphi^8 \rho_{\text{crit}}$  is

$$\text{KAL}_{\text{eff}}^2 = \frac{5\varphi^8}{2\varphi^4} = \frac{5}{2}\varphi^4 = \frac{15}{2}\alpha, \quad \text{KAL}_{\text{eff}} = \varphi^2 \sqrt{\frac{5}{2}}. \quad (36)$$

Note that  $\text{KAL}_{\text{eff}}^2 = (5/2)\varphi^4$  is indeed purely a power of  $\varphi$  (no  $\pi$ ), confirming the postulate is self-consistent. Substituting back into Eq. (34):

$$M^4 = 6\alpha \times \frac{15}{2}\alpha = 45\alpha^2 \rho_{\text{crit}} = 5\varphi^8 \rho_{\text{crit}}. \quad (37)$$

This is precisely Conjecture C.1.

The Route A discrepancy is now transparent:  $\text{KAL}_0/\text{KAL}_{\text{eff}} = [(\varphi + 3\pi)/2]/[\varphi^2 \sqrt{5/2}]$ , which involves  $\pi$  and equals  $\sqrt{1.779} \approx 1.334$  numerically. This ratio is the field-redefinition factor — the “missing”  $\pi$ -dependent normalisation between the inflationary and dark-energy kinematic bases.

**Conditional Theorem C.1.** *If the effective kinetic normalisation  $\text{KAL}_{\text{eff}}$  at the inflaton-to-quintessence transition is polynomial in  $\varphi$  (the UV-IR separability postulate), then the unique solution to  $M^4 = 6\alpha \text{KAL}_{\text{eff}}^2$  consistent with the inflationary  $\alpha$ -attractor  $K_\alpha$  is*

$$M^4 = 45\alpha^2 \rho_{\text{crit}} = 5\varphi^8 \rho_{\text{crit}}. \quad (38)$$

*The UV ladder closes:  $\varphi \rightarrow \alpha = \varphi^4/3 \rightarrow M^4 = 5\varphi^8 \rho_{\text{crit}} \rightarrow \alpha_K^{\text{full}} = 0.41691 \rightarrow H_0^{\text{UV}} = 73.040 \text{ km/s/Mpc}$  (0.00 $\sigma$  from *SH0ES*; conditional on Postulate C.1), entirely within the  $\{\varphi, \pi\}$  algebraic structure of *SSEE*.*

The unconditional proof requires an explicit derivation of  $\text{KAL}_{\text{eff}}$  from the  $P(X, \phi)$  Lagrangian at the inflationary transition — specifically, showing that the field-space Jacobian  $\partial\phi/\partial\chi|_{\text{transition}}$  evaluates to  $\varphi^2 \sqrt{5/2}$  rather than  $\text{KAL}_0 = (\varphi + 3\pi)/2$ . This is deferred to Paper B of the *SSEE* series.

## 7 Additional UV Predictions

Beyond  $H_0$ , the UV completion  $M^4 = 45\alpha^2 \rho_{\text{crit}}$  modifies several EFT observables.

**Kineticity  $\alpha_K$ .** The UV-corrected kineticity  $\alpha_K^{\text{full}} = 0.41691$  is in principle measurable by spectroscopic surveys that probe modified-gravity EFT parameters. Current constraints from DESI DR2 BAO [22] set the background context; forthcoming Euclid and DESI Y5 data will probe  $\alpha_K$  at the  $\sim 1\%$  level, distinguishing the UV-completed theory from the IR limit by  $3.4\sigma_{\text{sys}}$ .

**Effective equation of state.** The UV term  $X^2/M^4$  in  $K(X)$  modifies the effective pressure as

$$p_\phi^{\text{UV}} = \frac{X}{\text{KAL}_0} + \frac{X^2}{M^4} - V(\phi), \quad \rho_\phi^{\text{UV}} = \frac{X}{\text{KAL}_0} + \frac{3X^2}{M^4} + V(\phi), \quad (39)$$

Under the background matching of Eq. (10) — which holds  $\rho_\phi$  and  $\rho_\phi(1+w_0)$  fixed at their IR values —  $w_0$  is unchanged by construction (the field value  $X_{\text{bg}}$  adjusts instead). A residual shift appears only if one holds the field configuration  $(X_{\text{bg}}, V)$  fixed when switching on the UV term, in which case the first-order estimate is

$$\Delta w_0^{\text{UV}} = w_0^{\text{UV}} - w_0^{\text{IR}} \approx \frac{(1-3w_0)X_{\text{bg}}^2/M^4}{\rho_\phi} \approx +2.4 \times 10^{-3}. \quad (40)$$

Either way,  $|\Delta w_0| \lesssim 0.002$  lies well below the projected Euclid sensitivity ( $\sigma_{w_0} \approx 0.01$ ): the UV term leaves the background equation of state observationally indistinguishable from the IR limit, and the discriminating observable remains  $\alpha_K$ .

**Sound speed.** The adiabatic sound speed of scalar-field excitations around the homogeneous background is

$$c_{s,\text{ad}}^2 = \frac{K_X}{K_X + 2XK_{XX}} = \frac{1/\text{KAL}_0 + 2X/M^4}{1/\text{KAL}_0 + 6X/M^4} \approx 0.967. \quad (41)$$

This quantity governs how fast small adiabatic excitations of  $\phi$  propagate on top of the smooth dark-energy background. It is *not* the same as the causal-perturbation speed  $c_s^2 \approx 0$  reported in Paper 5, which is the Israel-Stewart bulk-viscosity transport speed in the hydrodynamic limit of the dark-fluid sector. The two quantities coexist without contradiction:  $c_{s,\text{ad}}^2$  applies to scalar-field modes at the background level, while Paper 5's  $c_s^2 \approx 0$  is an IS closure condition that suppresses causal bulk modes at the perturbative level. The 3.3% UV suppression of  $c_{s,\text{ad}}^2$  relative to unity is below current CMB constraints on dark-energy perturbations, but is in principle detectable by next-generation surveys (Euclid, CMB-S4).

## 8 Consistency with the SSEE Suite

Table 4 summarises how the UV completion is consistent with the nine preceding papers.

Table 4: Consistency of the UV completion with the SSEE suite.

| Paper        | Result used                                                    | UV impact                                                               |
|--------------|----------------------------------------------------------------|-------------------------------------------------------------------------|
| Paper 1      | $\alpha = \varphi^4/3 \rightarrow r = 0.00813$                 | <b>Direct input:</b> $M^4 = 45\alpha^2\rho_{\text{crit}}$               |
| Paper 5      | IS: $c_s^2 \approx 0$ (bulk-viscosity mode)                    | Compatible: distinct from $c_{s,\text{ad}}^2 \approx 0.967$ (field mod) |
| Paper 7      | $\alpha_K^{\text{IR}} = 0.4033$                                | Base for UV correction (+3.37%)                                         |
| Paper 8      | $\mathcal{M} = \mathcal{A}/2$ ; Vainshtein $r_V \propto 1/M^2$ | $r_V$ enlarged to $\sim 10^{44}$ m at UV; double GR protec              |
| Paper 9      | $H_0^{\text{IR}} = 72.86$ (1.12 $\sigma$ )                     | UV closes to 73.040 (0.00 $\sigma$ ; conditional <sup>a</sup> )         |
| Papers 2–4,6 | CMB, MCMC, $\sigma_8$ , $f\sigma_8$                            | $M$ only modifies $\alpha_K$ ; the algebraic constant $\mathcal{M} =$   |

<sup>a</sup> Canonical UV result  $H_0^{\text{UV}} = 73.040$  km/s/Mpc (0.00 $\sigma$  from SH0ES), conditional on Postulate C.1 (UV-IR separability, §6.3). It agrees with the SH0ES central value 73.04 to rounding; the absolute-scale origin (Postulate D) remains open, as in  $\Lambda$ CDM.

The AURA cancellation of Paper 9 ( $\mathcal{A}$  drops out of  $\alpha_K/(3\mathcal{M})$  in the IR limit) is technically broken in the UV because  $\alpha_K^{\text{full}}$  acquires an AURA-dependent correction. The canonical UV result  $H_0^{\text{UV}} = 73.040$  km/s/Mpc (0.00 $\sigma$  from SH0ES), conditional on Postulate C.1, indicates that the UV correction is a consistent completion rather than a destabilisation of the IR identity. It agrees with the SH0ES central value to within 0.001 km/s/Mpc. As stressed throughout, the absolute dimensional scale (Postulate D) is not an independent prediction — both are self-consistency checks.

## 9 Summary and Conclusions

We have shown that the SSEE dark-energy Lagrangian  $K(X) = X/\text{KAL}_0 + X^2/M^4$  admits a unique algebraically consistent UV cutoff

$$M^4 = 45\alpha^2\rho_{\text{crit}} = 5\varphi^8\rho_{\text{crit}} \approx 234.9\rho_{\text{crit}}, \quad (42)$$

where  $\alpha = \varphi^4/3$  is the inflationary  $\alpha$ -attractor parameter of Paper 1. With this cutoff, the full Bellini–Sawicki kineticity is  $\alpha_K^{\text{full}} = 0.41691$  and the canonical local Hubble constant is  $H_0^{\text{UV}} = 73.040$  km/s/Mpc ( $0.00\sigma$  from SH0ES), conditional on Postulate C.1, matching the SH0ES central value 73.04 to rounding.

The key conclusions are:

1. The algebraic identity  $45\alpha^2 = 5\varphi^8$  (exact,  $|\text{diff}| = 0$ ) connects the inflationary curvature  $\alpha$  to the dark-energy UV scale.
2. The UV correction to  $\alpha_K$  is perturbative ( $\varepsilon \approx 5.7 \times 10^{-4}$ ) and shifts  $\alpha_K$  by +3.37%.
3. The canonical IR Hubble residual of Paper 9 ( $H_0^{\text{IR}} = 72.86$ ,  $0.17\sigma$ ) is reduced to  $H_0^{\text{UV}} = 73.040$  ( $0.00\sigma$  from SH0ES) by the UV completion, conditional on Postulate C.1 (§6.3).
4. The same  $\alpha$  predicts  $r \approx 0.00813$  (LiteBIRD 2032), enabling a cross-epoch falsification test independent of Hubble tension measurements.
5. A first-principles derivation of  $M^4 = 45\alpha^2\rho_{\text{crit}}$  remains under investigation (Section 6), with Route C (full  $P(X, \phi)$  Lagrangian) identified as the most promising path.

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## References

- [1] M. E. Almeida Vallejo, *SSEE-V3.6: Algebraic Framework, EFT, and  $\alpha$ -Attractor Inflation*, SSEE Paper 1 (2026).
- [2] M. E. Almeida Vallejo, *Canonical EFT of SSEE:  $\beta_c = -\mathcal{A}$ ,  $\alpha_K = 0.4033$* , SSEE Paper 7 (2026).
- [3] M. E. Almeida Vallejo, *Strong-Gravity Regime: Disformal Geodesics and  $\mathcal{M}$  Lensing*, SSEE Paper 8 (2026).
- [4] M. E. Almeida Vallejo, *Hubble Tension via Algebraic Screening Fraction:  $f_{\text{screen}} = \alpha_K/(3\mathcal{M}) = (\pi - \varphi)/(\varphi + \pi)^2$* , SSEE Paper 9 (2026).
- [5] A. G. Riess et al., *A Comprehensive Measurement of the Local Value of the Hubble Constant*, ApJL **934**, L7 (2022) [arXiv:2112.04510].
- [6] W. L. Freedman et al., *The Carnegie-Chicago Hubble Program. IX. Calibration of the Tip of the Red Giant Branch Method in the Near-Infrared and the Value of the Hubble Constant*, ApJ **963**, 84 (2024) [arXiv:2308.02474]. [ $H_0 = 69.96 \pm 1.05$  (stat)  $\pm 1.12$  (sys) km/s/Mpc].
- [7] E. Bellini & I. Sawicki, *Maximal freedom at the cost of fine tuning in EFT of dark energy*, JCAP **07** (2014) 050 [arXiv:1404.3625].
- [8] LiteBIRD Collaboration, *LiteBIRD: a Satellite for the Studies of B-Mode Polarization and Inflation from Cosmic Background Radiation Detection*, JLT **200** (2020) 422 [arXiv:2202.02773].
- [9] A. A. Starobinsky, *A new type of isotropic cosmological models without singularity*, Phys. Lett. B **91** (1980) 99.
- [10] R. Kallosh & A. Linde, *Universality Class in Conformal Inflation*, JCAP **07** (2013) 002 [arXiv:1306.5220].
- [11] M. Galante, R. Kallosh, A. Linde & D. Roest, *Unity of Cosmological Inflation Attractors*, Phys. Rev. Lett. **114** (2015) 141302 [arXiv:1412.3797].

- [12] A. R. Liddle & S. M. Leach, *How long before the end of inflation were observable perturbations produced?*, Phys. Rev. D **68** (2003) 103503 [arXiv:astro-ph/0305263].
- [13] Planck Collaboration (Y. Akrami et al.), *Planck 2018 results. X. Constraints on inflation*, A&A **641**, A10 (2020) [arXiv:1807.06211].
- [14] K. N. Abazajian et al. (CMB-S4 Collaboration), *CMB-S4 Science Book, First Edition*, arXiv:1610.02743 (2016).
- [15] J. Garriga & V. F. Mukhanov, *Perturbations in k-inflation*, Phys. Lett. B **458** (1999) 219 [arXiv:hep-th/9904176].
- [16] G. 't Hooft, *Naturalness, chiral symmetry, and spontaneous chiral symmetry breaking*, in *Recent Developments in Gauge Theories*, NATO ASI Series B **59** (1980) 135.
- [17] S. Perlmutter et al., *Measurements of  $\Omega$  and  $\Lambda$  from 42 high-redshift supernovae*, ApJ **517**, 565 (1999) [arXiv:astro-ph/9812133].
- [18] A. G. Riess et al., *Observational evidence from supernovae for an accelerating universe and a cosmological constant*, AJ **116**, 1009 (1998) [arXiv:astro-ph/9805200].
- [19] L. Verde, T. Treu & A. G. Riess, *Tensions between the early and the late Universe*, Nat. Astron. **3**, 891 (2019) [arXiv:1907.10625].
- [20] M. Kamionkowski & A. G. Riess, *The Hubble tension and early dark energy*, Annu. Rev. Nucl. Part. Sci. **73**, 153 (2023) [arXiv:2211.04492].
- [21] Planck Collaboration, *Planck 2018 results VI: Cosmological parameters*, A&A **641**, A6 (2020) [arXiv:1807.06209].
- [22] M. Abdul-Karim et al. (DESI Collaboration), *DESI DR2 BAO measurements*, arXiv:2503.14738 (2025).
- [23] E. J. Copeland, M. Sami & S. Tsujikawa, *Dynamics of dark energy*, Int. J. Mod. Phys. D **15**, 1753 (2006) [arXiv:hep-th/0603057].
- [24] D. H. Weinberg et al., *Observational probes of cosmic acceleration*, Phys. Rep. **530**, 87 (2013) [arXiv:1201.2434].
- [25] G. W. Horndeski, *Second-order scalar-tensor field equations in a four-dimensional space*, Int. J. Theor. Phys. **10**, 363 (1974).
- [26] G. Gubitosi, F. Piazza & F. Vernizzi, *The effective field theory of dark energy*, JCAP **2013**, 032 (2013) [arXiv:1210.0201].
- [27] C. Armendariz-Picon, V. Mukhanov & P. J. Steinhardt, *Essentials of k-essence*, Phys. Rev. D **63**, 103510 (2001) [arXiv:astro-ph/0006373].
- [28] C. Cheung, P. Creminelli, A. L. Fitzpatrick, J. Kaplan & L. Senatore, *The effective field theory of inflation*, JHEP **2008**, 014 (2008) [arXiv:0709.0293].
- [29] M. Chevallier & D. Polarski, *Accelerating universes with scaling dark matter*, Int. J. Mod. Phys. D **10**, 213 (2001) [arXiv:gr-qc/0009008].
- [30] E. V. Linder, *Exploring the expansion history of the universe*, Phys. Rev. Lett. **90**, 091301 (2003) [arXiv:astro-ph/0208512].
- [31] R. R. Caldwell, *A phantom menace? Cosmological consequences of a dark energy component with super-negative equation of state*, Phys. Lett. B **545**, 23 (2002) [arXiv:astro-ph/9908168].
- [32] B. Ratra & P. J. E. Peebles, *Cosmological consequences of a rolling homogeneous scalar field*, Phys. Rev. D **37**, 3406 (1988).